

Computer Networks Masterclass

OSI Model, TCP/IP, Routing, Switching & Network Protocols

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Module 1: Introduction to Computer Networks

1. What is a Computer Network?

- **Definition:** A collection of autonomous computing devices interconnected by communication channels to share data, hardware resources, and services.
- **Key Objectives:** Resource sharing, high reliability, scalability, and cost reduction.
- **Basic Components:** Sender, receiver, message, transmission medium (cables, fiber, wireless), and protocols.

2. Types of Networks by Geographic Scale

- **PAN (Personal Area Network):** Centered around an individual in a single room (e.g., Bluetooth devices).
- **LAN (Local Area Network):** Connects computers within a limited area like a home, office, or school campus.
- **MAN (Metropolitan Area Network):** Spans an entire city or town (e.g., urban cable network).
- **WAN (Wide Area Network):** Connects computers across large physical distances, countries, or continents (The Internet).

3. Network Topologies

- **Bus Topology:** Single central backbone cable; simple layout but prone to single-point failure.
- **Star Topology:** All devices connected to a central hub/switch; easy management and robust against individual cable failures.
- **Ring Topology:** Devices connected in a circular loop; data travels in one direction.
- **Mesh Topology:** Every device connected to multiple or all other devices; offers high redundancy and fault tolerance.

Module 2: The OSI Reference Model

1. Introduction to the OSI Model

A conceptual 7-layer framework developed by ISO to standardize network communications using a modular "divide and conquer" philosophy.

- **Layer 1 (Physical):** Electrical voltages, raw bit streams, physical medium and cables.
- **Layer 2 (Data Link):** Node-to-node transfer, framing, MAC addressing, error detection.
- **Layer 3 (Network):** Packet routing across independent networks, logical IP addressing.
- **Layer 4 (Transport):** Reliable end-to-end data delivery, flow control, multiplexing (TCP, UDP).
- **Layer 5-7 (Session, Presentation, Application):** Dialog control, encryption, data translation, and user-facing network applications.

Module 3: The TCP/IP Protocol Suite

1. Architecture & Encapsulation

- **Four-Layer Model:** Network Access, Internet, Transport, and Application layers powering the modern Internet.
- **Encapsulation / Decapsulation:** Control headers are added as data moves downward (Application → Segment → Packet → Frame → Bits) and stripped sequentially upon receipt.

Module 4: Switching & Data Link Protocols

- **Ethernet & MAC Addressing:** Standard wired LAN technology (IEEE 802.3) using 48-bit physical MAC identifiers and collision detection mechanisms (CSMA/CD).
- **Switching Fundamentals:** Layer 2 switches inspect frame MAC addresses and forward data selectively using dynamic CAM tables. VLANs provide logical broadcast domain segmentation.

Module 5: Network Layer & Routing

- **IP Addressing:** IPv4 uses 32-bit dotted-decimal addresses (supported by subnetting and CIDR), while IPv6 uses a 128-bit hexadecimal format to overcome address exhaustion.
- **Routing Protocols:**
 - **Distance Vector (RIP):** Uses hop counts and the Bellman-Ford algorithm.
 - **Link-State (OSPF):** Builds complete topological maps using Dijkstra's algorithm.
 - **Path-Vector (BGP):** Manages inter-autonomous system communication across the Internet core.

Module 6: Transport Layer Protocols

- **TCP (Transmission Control Protocol):** Connection-oriented, reliable protocol utilizing a 3-way handshake (SYN, SYN-ACK, ACK), ordered sequencing, flow control, and congestion avoidance.
- **UDP (User Datagram Protocol):** Connectionless, unreliable protocol prioritizing transmission speed over guaranteed delivery (ideal for streaming, VoIP, DNS).

Module 7: Application Layer & Security

Protocol	Function & Purpose
HTTP / HTTPS	Transfers secure web documents over the World Wide Web.
DNS	Translates domain names to numerical IP addresses (Port 53).
DHCP	Automatically assigns IP addresses and configurations to client devices.
SMTP / IMAP	Handles outgoing mail transmission and incoming mail retrieval.

Network Security Fundamentals

Protection is enforced via firewalls (traffic filtering), VPNs (encrypted tunnels over public networks), and cryptographic protocols (SSH, TLS, IPsec) to secure data in transit.